Deep learning: Portfolio assignment 4
Jeffrey Gerrits - Master of Applied Data Science UOS3 - 15-03-2026

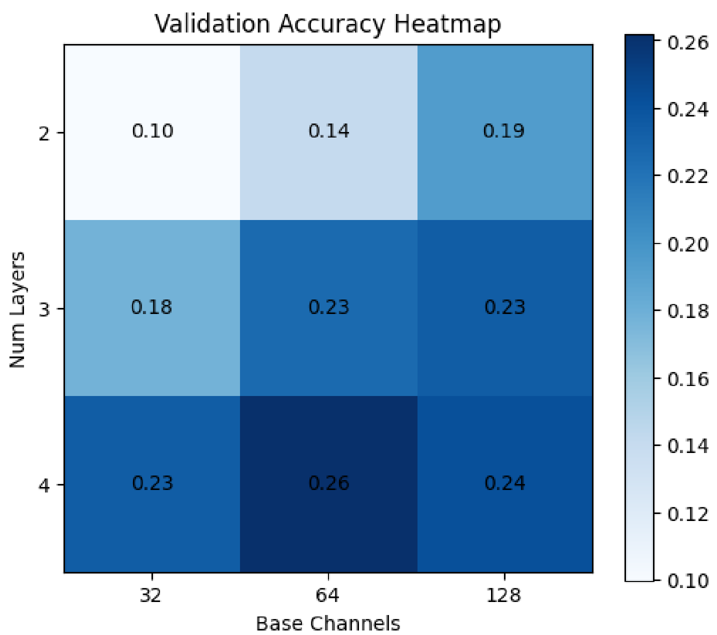
For this experiment a convolutional neural network (CNN) was trained on the Oxford Flowers102 dataset by using Ray Tune. The Flowers102 dataset contains 102 flower classes with 40 to 258 images per class. Transfer learning was initially considered, but this resulted in memory errors on the available VM. Therefore a simpler CNN architecture was used. Because the dataset contains many classes relative to the number of images, learning useful visual features is challenging for a small CNN. However, the CNN provided a good basis for making improvements through hyperparameter tuning with Ray Tune.

Iteration 1

The first iteration focused on basic network architecture. To speed up testing, a small image size (96x96) and a small number of epochs (6) was used. It was expected that, given a relatively small dataset, a deep network performs better than a wide network. To see if both a deep and a wide network performed equally well (null hypothesis) or that there is a difference (alternative hypothesis), Ray Tune was used to systematically explore combinations of hyperparameters. A grid search approach was used to test variations in network width, network depth and learning rate. Ray tested multiple configurations and tracked multiple metrics for each trial, in particular the validation accuracy.

The results showed a large variation in validation accuracy between the different trials, ranging from 0.06 to 0.28. This confirms that the choice of hyperparameters strongly affects performance. The deeper models, with a higher number of layers (e.g. 4 layers and 64 base channels), consistently performed better than shallow ones (e.g. 2 layers and 128 base channels). Therefore the null hypothesis of this first iteration was rejected.

Figure 1 Validation Accuracy for Different Combinations of Model Depth and Width During Iteration 1



Another useful outcome of this first iteration was identifying promising regions of the search space. These results were used as input for a second iteration of hyperparameter tuning.

Iteration 2

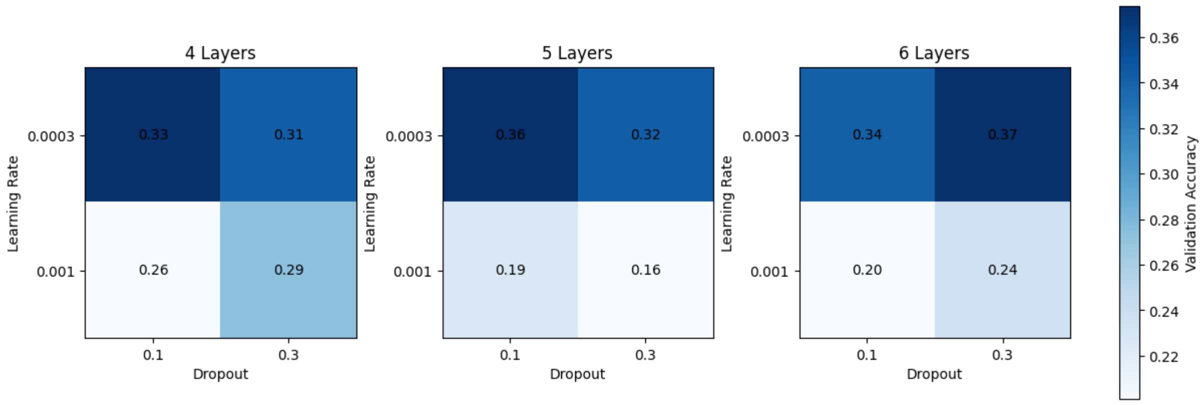
In the second iteration Ray was again used to explore a refined search space with different combinations of hyperparameters. The image size and number of epochs was increased to 128x128 pixels and 10 epochs.

The expectation for this iteration was that when networks become deeper, regularization hyperparameters and the learning rate become more important in relation to improving the network. The hypothesis states:
H0: If the depth of a model is increased, the effect of regularization and the learning rate does not change.
H1: If the depth of a model is increased, regularization and the learning rate increasingly affect the performance of the model.

To investigate the effects of these hyperparameters, first the model width was fixed at 64 base channels. The search space configuration used by Ray was set up with grid searches for:

- Drop out values of 0.1 and 0.3.
- Leaning rates of 3e-4 and 1e-3.
- A number of layers between the range of 4 to 7. Increasing this hyperparameter, increases the depth of the model.

The results of the 12 trials of this iteration showed that deeper networks consistently improved validation accuracy, with the best configuration achieving 0.37 validation accuracy. This is a clear improvement compared to the best results of the first iteration (0.27). However, in all trials the model overfitted the training data. The two best performing trials scored 0.68 and 0.79 training accuracy, but only managed to score a validation accuracy of 0.37 and 0.34.



The null hypothesis was rejected based on two patterns that were observed in the results. First, higher dropout values (0.3) improved validation performance for deeper models, suggesting that regularization becomes more important as model capacity increases. Second, a lower learning rate (0.003) resulted in more stable training behavior, while higher learning rates often resulted in worse validation performance. We therefore conclude that, if the depth of a model is increased, regularization and the learning rate increasingly affect the performance of the model.

Reflection

Although the model outperforms random guessing for 102 classes, I am aware that the results of this (overfitting) CNN model are poor. The experiments were mainly useful as a learning experience in hyperparameter tuning with Ray Tune. It was also interesting to work with running time consuming experiments in .py scripts and managing them on a VM using tmux. For future experiments it would probably be more efficient to perform the first tuning iterations in a GPU environment such as Google Colab, in order to narrow down the search space faster before running longer experiments on the VM.

Appendix A - Configuration settings Ray Tune

Iteration 1

```
config = {
    "data_dir": str(data_dir),
    "lr": tune.grid_search([3e-4, 1e-3]),
    "base_channels": tune.grid_search([32, 64, 128]),
    "num_layers": tune.grid_search([2, 3, 4]),
    "dropout": 0.2,
  }
```

Iteration 2

```
  config = {
    "data_dir": str(data_dir),
    "lr": tune.grid_search([3e-4, 1e-3]),
    "base_channels": 64,
    "num_layers": tune.grid_search(range(4, 7)),
    "dropout": tune.grid_search([0.1, 0.3]),
  }
```

Appendix B - Results hyperparameter tuning

Iteration 1

trial_id	learning rate	base channels	num_layers	dropout	train accuracy	val accuracy	val loss
cd65b_00000	0.0003	32	2	0.2	0.05	0.07	4.34
cd65b_00001	0.0003	64	2	0.2	0.10	0.10	4.15
cd65b_00002	0.0003	128	2	0.2	0.16	0.19	3.89
cd65b_00003	0.0010	32	2	0.2	0.10	0.13	4.01
cd65b_00004	0.0010	64	2	0.2	0.13	0.18	3.77
cd65b_00005	0.0010	128	2	0.2	0.20	0.19	3.52
cd65b_00006	0.0003	32	3	0.2	0.12	0.16	4.05
cd65b_00007	0.0003	64	3	0.2	0.20	0.22	3.71
cd65b_00008	0.0003	128	3	0.2	0.28	0.25	3.40
cd65b_00009	0.0010	32	3	0.2	0.17	0.20	3.66
cd65b_00010	0.0010	64	3	0.2	0.23	0.23	3.37
cd65b_00011	0.0010	128	3	0.2	0.24	0.22	3.25
cd65b_00012	0.0003	32	4	0.2	0.24	0.25	3.49
cd65b_00013	0.0003	64	4	0.2	0.33	0.28	3.14
cd65b_00014	0.0003	128	4	0.2	0.32	0.27	3.02
cd65b_00015	0.0010	32	4	0.2	0.22	0.22	3.24
cd65b_00016	0.0010	64	4	0.2	0.27	0.25	3.12
cd65b_00017	0.0010	128	4	0.2	0.26	0.21	3.30

Iteration 2

trial_id	learning rate	base channels	num_layers	dropout	train accuracy	val accuracy	val loss
0ffef_00000	0.0003	64	4	0.1	0.43	0.33	2.86
0ffef_00001	0.0003	64	4	0.3	0.39	0.31	2.94
0ffef_00002	0.0010	64	4	0.1	0.41	0.26	3.01
0ffef_00003	0.0010	64	4	0.3	0.36	0.29	2.96
0ffef_00004	0.0003	64	5	0.1	0.58	0.36	2.57
0ffef_00005	0.0003	64	5	0.3	0.51	0.32	2.71
0ffef_00006	0.0010	64	5	0.1	0.39	0.19	4.42
0ffef_00007	0.0010	64	5	0.3	0.36	0.16	4.09
0ffef_00008	0.0003	64	6	0.1	0.79	0.34	2.81
0ffef_00009	0.0003	64	6	0.3	0.68	0.37	2.48
0ffef_00010	0.0010	64	6	0.1	0.28	0.20	3.31
0ffef_00011	0.0010	64	6	0.3	0.33	0.24	3.40